UNIVERSITY OF LONDON

BSc EXAMINATION 2025

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

IY3609: Digital Forensics

Time Allowed: **TWO hours**

Attempt **THREE** questions
Please answer each question on a separate page
No credit will be given for attempting any further questions

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Question 1: Storage Forensics 1

(a) i. How can a forensic analyst confirm that a disk uses the Master Boot Record (MBR) partitioning scheme? Provide two observations from the MBR structure. [2 marks]

ii. When examining a disk with a single unknown file system, briefly describe two approaches a forensic analyst might take to identify the file system. [2 marks]

iii. Briefly explain which information the FSINFO structure in a FAT32 file system contains and what a forensic analyst can infer from that. [2 marks]

(b) Catherine is analysing an image of a FAT32 file system. The file system contained a text file named `notes.txt` that was recently deleted. Using a hex editor, she identifies and examines the directory entry for the deleted file and finds the following information:

- First cluster (MSB): `0x0000`
- First cluster (LSB): `0x002A`
- File size: 5500 bytes

The entry in the FAT table for the first cluster is `0x00000000`. Catherine now wants to determine if the file can be recovered.

i. Determine the address of the first cluster in decimal representation. [2 marks]

ii. Describe three steps Catherine can take to attempt full recovery of the deleted file. [3 marks]

iii. The cluster size of the file system is 4 KiB. The first cluster starts with the following hexadecimal data: `54 68 69 73 20 69 73 20 74 68 65`

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`20 63 6F 6E 74 ...`, which corresponds to the ASCII representation "This is the cont ...". Subsequent clusters contain data that looks random in ASCII representation. Quantify the amount of data in bytes that Catherine *cannot* easily recover. Justify your answer. [4 marks]

(c) Carlos is tasked with recovering data from a forensic duplicate obtained from a storage appliance. The appliance uses a combination of FAT32 and a proprietary data deduplication algorithm to optimize storage space.

Discuss two challenges Carlos may face when trying to analyse the image file with standard forensic software. How could Carlos overcome these challenges? [10 marks]

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Question 2: Storage Forensics 2

(a) i. Briefly explain how access to a volume shadow copy (as created by the Windows Volume Shadow Copy Service) can assist forensic analysts in reconstructing deleted or modified files. [2 marks]

ii. Why is understanding the RAID configuration of a storage array critical for forensic analysis? [2 marks]

iii. Briefly describe two challenges that NTFS self-healing mechanisms may pose for forensic analysts. [2 marks]

(b) Emma, a forensic analyst, has been tasked with recovering data from a storage server seized in a high-profile corporate fraud case. The storage server contains a set of SSD drives in a RAID 5 configuration. Together the SSDs form a logical volume that is protected with Microsoft BitLocker encryption. During her analysis, Emma discovers that one drive in the RAID array is missing. Furthermore, the BitLocker key is not immediately available.

i. Explain how the redundancy mechanism of RAID 5 can assist in recovering data from the array, even when one drive is missing. [3 marks]

ii. Suggest two approaches Emma could take to obtain plaintext access to the encrypted data. For each approach, outline two required conditions or assumptions for it to succeed. [6 marks]

(c) Alex is assisting a company after a flood destroyed parts of their main storage server. The storage server has an 8-disk RAID 10 configuration as shown below.

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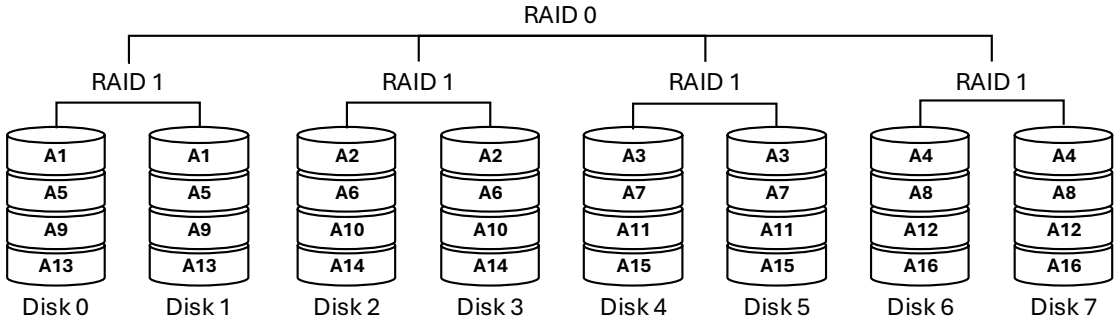

Figure 1: RAID 10 configuration

i. Assume disks 1, 4, and 5 were completely destroyed. Is it still possible to fully recover all data from the remaining disks? If yes, provide a detailed reasoning why. If not, argue whether it's possible to at least partially recover data. [6 marks]

ii. How does this RAID 10 configuration compare to an 8-disk RAID 5 in terms of fault tolerance? As part of your answer, state how many disk failures each configuration can tolerate at most. [4 marks]

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Question 3: Host Forensics

(a) i. Briefly describe two challenges forensic analysts face when performing live forensics on a running system. [2 marks]

ii. Briefly explain why inconsistencies can occur in memory snapshots during in-system memory acquisition and discuss their impact on forensic investigations. [2 marks]

iii. Describe one advantage and one disadvantage of using external hardware-based tools for memory acquisition. [2 marks]

(b) Sasha is investigating a live Windows 11 system that exhibits unusual network behaviour. She decides to examine the open ports and active connections to identify signs of potential compromise.

i. Explain two ways Sasha can identify open ports and active connections on the live system. [4 marks]

ii. Sasha notices that some network connections are in the `CLOSE_WAIT` state. Explain what this state indicates in the TCP connection lifecycle and why it might be of forensic interest. [6 marks]

(c) David is a security investigator analysing a virtual Windows 11 machine hosted on his company's VMware server. The company suspects that an employee had temporarily installed expensive software without a license to complete a difficult task. As the employee denies this allegation, David is tasked with finding evidence.

Discuss three artefacts David can analyse from the VM's virtual disk to identify signs of unauthorized software installations. [9 marks]

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Question 4: Malware Forensics

(a) i. Define the term polymorphic malware and explain how it differs from metamorphic malware in its method of avoiding detection. [2 marks]

ii. Describe two challenges that document-based malware poses to forensic analysts. [2 marks]

iii. Explain the purpose of a sandbox environment in malware analysis and discuss one limitation of this approach when analysing sophisticated malware. [2 marks]

(b) David is inspecting a live corporate workstation suspected of being infected with malware that exfiltrates sensitive files to an external server.

i. Describe two methods David can use to analyse network activity generated by the malware. [4 marks]

ii. David identifies an unknown executable on the workstation that he suspects is malware. Describe two methods David can use to perform static analysis of the executable. [4 marks]

iii. How might code obfuscation complicate David's investigation of the executable? [1 marks]

(c) Sophia is investigating an advanced malware sample discovered on a corporate network. The malware appears to execute in stages and utilizes techniques to detect when it is being run in a controlled environment.

i. Sophia decides to perform dynamic analysis to investigate the malware's behaviour. Briefly describe two precautions she should take to ensure that the malware cannot cause harm during the analysis. [6 marks]

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ii. Sophia observes that the malware downloads an encrypted payload during execution. Briefly discuss one challenge this poses to her analysis and one potential method to overcome it. [4 marks]

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Question 5: Mobile Forensics

(a) i. Give two examples of artefacts related to user activity that can be retrieved during forensic analysis from data stored by mobile applications on Android and iOS phones. [2 marks]

ii. Briefly explain what is meant by chip-off analysis in mobile forensics and discuss one major limitation of this method for modern smartphones. [2 marks]

iii. Modern smartphones often include secure enclaves for sensitive data. Briefly explain the role of a secure enclave and why it complicates forensic investigations. [2 marks]

(b) Liam is a forensic investigator tasked with extracting the chat history from a messaging app on a locked iPhone 15 running iOS 18. The iPhone is protected with Face ID and a six-digit passcode. The device is configured to create regular iCloud backups, which may include app data.

i. Suggest three ways Liam can attempt to access the iPhone. Briefly describe each approach and state one key requirement for it to be successful. [6 marks]

ii. Assume Liam cannot unlock the iPhone. Describe how he can attempt to access and analyse its iCloud backup instead to obtain the messaging app's chat history. Also discuss one key requirement for this method to be successful and one potential limitation. Assume that the Advanced Data Protection feature is not enabled for the iCloud backup. [4 marks]

(c) Tabby is a forensic investigator tasked with determining whether a suspect used their iPhone to photograph confidential documents at a specific time

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and location. During the investigation, Tabby discovers that the iPhone is jailbroken and has a suspicious application installed, which may interfere with the device's integrity.

i. Propose two approaches Tabby could use to determine whether the suspicious application manipulated the photos or their metadata. [6 marks]

ii. Which counter-forensic techniques might the suspicious application use to hinder Tabby's analysis? Describe one example. [3 marks]

END

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